Section C Comprehensive Review – Mixed

Directions: Show all work. Use exact values. Box or circle all answers.

1. Point: $(4, 81)$; common ratio = 3. Write two different explicit equations that represent the function.

2. For $y = f(x) = (750)(1.12)^x$, identify the y-intercept and common ratio. At what percentage is the function growing or decaying? Is it growth or decay?

3. y-intercept = -10; common ratio = $\frac{1}{4}$. Write an explicit equation.

4. y-intercept = -3; common ratio = $\frac{2}{3}$. Write an explicit equation.

5. Convert the recursive formula to an explicit formula: $a_1 = -25, \quad a_n = -\frac{2}{5}(a_{n-1})$.

6. Convert the explicit formula to a recursive formula. Include the first term: $a_n = 11(4)^{n-1}$.

7. The table shows the number of cells in a culture for hours 1 through 4.

Hour	1	2	3	4
Cells	5	20	80	320

Find the common ratio, write an explicit formula, and find the value at hour 6.

8. A sample contains 640 grams when an experiment starts and decreases by 25% each period. Write an exponential equation to model the amount remaining at any given time. How many grams remain after 2 periods?

9. Point: $(3, 8)$; common ratio = 2. Write three different explicit equations that represent the function.

10. For $y = f(x) = (900)(0.65)^x$, identify the y-intercept and common ratio. At what percentage is the function growing or decaying? Is it growth or decay?

11. y-intercept = 8; common ratio = 6. Write an explicit equation.

12. Write an explicit formula for the geometric sequence $-4, 12, -36, 108, \ldots$.

13. Point: $(1, 18)$; common ratio = 3. Write two different explicit equations that represent the function.

14. Convert the explicit formula to a recursive formula. Include the first term: $a_n = 3(-5)^{n-1}$.

15. A bacteria culture has 200 cells when an experiment starts and increases by 50% each hour. Write an exponential equation to model the number of cells at any given time. How many cells are present after 3 hours?

16. y-intercept = 15; common ratio = $\frac{4}{5}$. Write an explicit equation.

17. y-intercept = 24; common ratio = 2. Write an explicit equation.

18. Point: $(2, 32)$; common ratio = 2. Write three different explicit equations that represent the function.

19. Point: $(2, \frac{3}{2})$; common ratio = $\frac{1}{2}$. Write three different explicit equations that represent the function.

20. Convert the recursive formula to an explicit formula: $a_1 = 8, \quad a_n = -2(a_{n-1})$.

21. Write an explicit formula for the geometric sequence $256, 128, 64, 32, \ldots$.

22. A geometric sequence has $a_3 = 28$ and $a_6 = 224$. Find the common ratio and write an explicit formula for the sequence.

23. Write an explicit formula for the geometric sequence $10, 30, 90, 270, \ldots$.

24. Convert the recursive formula to an explicit formula: $a_1 = 144, \quad a_n = \frac{1}{3}(a_{n-1})$.

25. Point: $(2, -12)$; common ratio = 2. Write three different explicit equations that represent the function.

26. The table shows the value of a device at the end of each year.

Year	1	2	3	4
Value ($)	2700	900	300	100

Find the common ratio, write an explicit formula, and find the value at year 5.

27. Convert the recursive formula to an explicit formula: $a_1 = 6, \quad a_n = 5(a_{n-1})$.

28. A machine is worth $\$5000$ when it is purchased and decreases by 10% each year. Write an exponential equation to model its value at any given time. What is its value after 2 years?

29. Convert the explicit formula to a recursive formula. Include the first term: $a_n = 128(\frac{1}{4})^{n-1}$.

30. A geometric sequence has $a_2 = -10$ and $a_5 = 80$. Find the common ratio and write an explicit formula for the sequence.

31. Write an explicit formula for the geometric sequence $192, 96, 48, 24, \ldots$.

32. Convert the explicit formula to a recursive formula. Include the first term: $a_n = -18(\frac{2}{3})^{n-1}$.